

The compact Brown–Zudilin weights: proofs for the companion rows

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Abstract

Let $T(n, k, l) = \binom{n+k}{n} \binom{n}{k}^2 \binom{n+l}{n} \binom{n}{l}^2 \binom{n+k+l}{n}$ and let $Q_n, \hat{P}_n, P_n$ be the coefficient sequences of the totally symmetric Brown–Zudilin cellular linear forms for $\zeta(5)$, so that $I_n = Q_n(2\zeta(5) + 4\zeta(3)\zeta(2)) - 4\hat{P}_n\zeta(2) - 2P_n$. We prove the compact closed form $\hat{P}_n = \sum_{k,l} T(n, k, l) \hat{w}_3(n, k, l)$ for the middle companion row, and we prove the exact Barnes-rational representation $P_n = -\frac{1}{2} \sum_{k,l} T(n, k, l) W^{(1)}(k, l)$ for the top row, where $W^{(1)}$ is an explicit polynomial in harmonic and finite Euler sums. Both proofs are uniform in n and use no $\mathbb{Q}$ -linear-independence assumptions about zeta values. The remaining step from the Barnes-rational representation to the compact three-term weight w_5 is a single identity between finite sums, which we isolate, reduce to a linear-algebra problem over a space of proved residue functionals, and discuss in the final section.

1 Statement of results

Throughout, $H_m^{(r)} = \sum_{i=1}^m i^{-r}$ (with $H_m^{(r)} = 0$ for $m \leq 0$), and

$$T(n, k, l) = \binom{n+k}{n} \binom{n}{k}^2 \binom{n+l}{n} \binom{n}{l}^2 \binom{n+k+l}{n}, \quad 0 \leq k, l \leq n.$$

Brown–Zudilin [1, Sec. 2] attach to the totally symmetric cellular integral

$$I_n = \int \cdots \int_{0 < t_1 < \cdots < t_5 < 1} \left(\frac{t_1(t_2-t_1)(t_3-t_2)(t_4-t_3)(t_5-t_4)(1-t_5)}{(t_3-t_1)t_3(1-t_4)(t_4-t_2)(t_5-t_2)} \right)^n \frac{dt_1 \cdots dt_5}{(t_3-t_1)t_3(1-t_4)(t_4-t_2)(t_5-t_2)}$$

the decomposition

$$I_n = Q_n(2\zeta(5) + 4\zeta(3)\zeta(2)) - 4\hat{P}_n\zeta(2) - 2P_n, \quad (1)$$

where $Q_n, \hat{P}_n, P_n$ are rational numbers. All four sequences satisfy one third-order Apéry-type recursion (produced by creative telescoping for I_n and checked directly for the rational sequences), with initial data

$$Q_0 = 1, Q_1 = 21, Q_2 = 2989, \quad \hat{P}_0 = 0, \hat{P}_1 = \frac{101}{4}, \hat{P}_2 = \frac{344923}{96}, \quad P_0 = 0, P_1 = \frac{87}{4}, P_2 = \frac{1190161}{384},$$

and (1) is *proved* in [1] by exactly this route: both sides satisfy the same recursion and agree at $n = 0, 1, 2$. In particular (1) is an identity between real numbers that does not presuppose any linear independence of zeta values; the sequences $Q_n, \hat{P}_n, P_n$ are pinned by the recursion and the initial data. Brown–Zudilin also prove

$$Q_n = \sum_{k,l=0}^n T(n, k, l), \quad (2)$$

again by checking the recursion.

Define, for $r \geq 1$,

$$A_r(x) = H_{n+x}^{(r)} - H_x^{(r)}, \quad B_r(x) = H_{n-x}^{(r)} - H_x^{(r)}, \quad C_r = H_{n+k+l}^{(r)} - H_{k+l}^{(r)},$$

$$\alpha = A_1(k) - A_1(l), \quad \beta = B_1(k) - B_1(l), \quad \Psi = \frac{\alpha}{2} + \beta,$$

and the *compact weights*

$$\hat{w}_3 = H_{n+k}^{(3)} - \Psi H_{n+k}^{(2)}, \quad w_5 = H_{n+k}^{(5)} + \frac{\alpha-\beta}{2} H_{n+k}^{(4)} + \left(\frac{A_2(k)+A_2(l)}{4} - \frac{\alpha\Psi}{2} \right) H_{n+k}^{(3)}.$$

Finally, let $L_k = -A_1(k) - C_1 - 2B_1(k)$ and $L_l = -A_1(l) - C_1 - 2B_1(l)$, and let $r_{11}, r_{12}, r_{21}, r_{22}$ be the explicit weight- ≤ 5 expressions in harmonic and finite Euler sums displayed in Proposition 3 below. Put

$$W^{(1)}(k, l) = r_{22}(k, l) + L_k r_{12}(k, l) + L_l r_{21}(k, l) + (L_k L_l - C_2) r_{11}(k, l). \quad (3)$$

The results of this paper are:

Theorem 1 (middle row). *For every $n \geq 0$,*

$$\hat{P}_n = \sum_{k,l=0}^n T(n, k, l) \hat{w}_3(n, k, l).$$

Theorem 2 (top row, Barnes-rational form). *For every $n \geq 0$,*

$$P_n = -\frac{1}{2} \sum_{k,l=0}^n T(n, k, l) W^{(1)}(k, l).$$

Theorem 3 (top row, compact form; conditional). *If the bridge identity*

$$\sum_{k,l=0}^n T(n, k, l) W^{(1)}(k, l) = -2 \sum_{k,l=0}^n T(n, k, l) w_5^{\text{sym}}(n, k, l) \quad (n \geq 0) \quad (4)$$

holds, where w_5^{sym} is the $k \leftrightarrow l$ symmetrisation of w_5 , then

$$P_n = \sum_{k,l=0}^n T(n, k, l) w_5(n, k, l) \quad (n \geq 0).$$

Identity (4) has been verified exactly for $n \leq 12$ and modulo two 31-bit primes for $n \leq 80$; Section 9 reduces it to membership of one explicit vector in the span of proved residue functionals and records the current state of that reduction. It is the only unproved statement in this paper.

Everything else below is a complete proof of Theorems 1 and 2. The logical skeleton is:

1. the Barnes kernel and its bivariate partial fractions (Sections 2–3);
2. the recombination $I_n = \sum T \cdot W_B$ over four universal sine-kernel integrals (Section 4);
3. the exact evaluation of the universal integrals in the basis $1, \zeta(2), \zeta(3), \zeta(4), \zeta(5), \zeta(2)\zeta(3)$ (Section 5);
4. four kernel identities: the $\zeta(5)$ and $\zeta(2)\zeta(3)$ coefficients equal $2Q_n$ and $4Q_n$; the $\zeta(4)$ and $\zeta(3)$ coefficients vanish; the $\zeta(2)$ coefficient equals $-4 \sum T w_3^{\text{sym}}$ (Section 6);

5. the middle row by descent through the generalised Beukers integral and Zudilin's evaluation lemma (Section 7);
6. the subtraction anchor: combining 1–5 with (1) kills every transcendental term exactly and leaves Theorem 2 (Section 8).

Remark 1. Step 6 is where previous attempts stalled, because comparing coefficients of $\zeta(2)$ or $\zeta(5)$ between two evaluations of I_n is illegitimate while the $\mathbb{Q}$ -linear independence of $1, \zeta(2), \zeta(3), \zeta(5), \zeta(2)\zeta(3)$ remains open. The subtraction route avoids this: once the middle row is proved *independently* (Theorem 1), the two evaluations of the real number I_n differ only in their rational terms, and equality of real numbers then forces equality of those rational terms. No independence is used anywhere.

2 The Barnes kernel

Brown–Zudilin's Barnes-type representation [1, eq. (intJ)], specialised to the totally symmetric parameters $p_0 = p_1 = p_2 = p_4 = p_5 = p_6 = q_1 = \dots = q_5 = n$, $p_3 = 2n$ (for which the normalising factorial prefactor collapses to $n!$), reads

$$I_n = \frac{n!}{(2\pi i)^2} \int_{-c_1-i\infty}^{-c_1+i\infty} \int_{-c_2-i\infty}^{-c_2+i\infty} \frac{\Gamma(n+1+s)^3 \Gamma(-s)}{\Gamma(2n+2+s)^2} \frac{\Gamma(n+1+t)^3 \Gamma(-t)}{\Gamma(2n+2+t)^2} \Gamma(2n+2+s+t) \Gamma(-n-1-s-t) ds dt, \quad (5)$$

valid for $0 < c_1, c_2 < n+1$ and $c_1 + c_2 > n+1$; we fix $c_1 = c_2 = n + \frac{2}{3}$.

Lemma 1. *With $c_1 = c_2 = n + \frac{2}{3}$,*

$$I_n = \frac{(-1)^n n!}{(2\pi i)^2} \iint R_n(s, t) \frac{\pi}{\sin \pi s} \frac{\pi}{\sin \pi t} \frac{\pi}{\sin \pi(s+t)} ds dt,$$

where

$$R_n(s, t) = \frac{\prod_{j=1}^n (s+j) \prod_{j=1}^n (t+j) \prod_{j=n+2}^{2n+1} (s+t+j)}{\prod_{j=n+1}^{2n+1} (s+j)^2 \prod_{j=n+1}^{2n+1} (t+j)^2}.$$

Proof. Apply the reflection formula $\Gamma(z)\Gamma(1-z) = \pi/\sin \pi z$ three times:

$$\Gamma(-s) = \frac{-\pi}{\sin(\pi s) \Gamma(1+s)}, \quad \Gamma(-t) = \frac{-\pi}{\sin(\pi t) \Gamma(1+t)},$$

and, using $\sin \pi(-n-1-u) = (-1)^n \sin \pi u$,

$$\Gamma(-n-1-s-t) = \frac{(-1)^n \pi}{\sin(\pi(s+t)) \Gamma(n+2+s+t)}.$$

The three sign factors multiply to $(-1)^n$. The remaining gamma quotients are rational:

$$\frac{\Gamma(n+1+s)^3}{\Gamma(2n+2+s)^2 \Gamma(1+s)} = \frac{\prod_{j=1}^n (s+j)}{\prod_{j=n+1}^{2n+1} (s+j)^2}, \quad \frac{\Gamma(2n+2+s+t)}{\Gamma(n+2+s+t)} = \prod_{j=n+2}^{2n+1} (s+t+j). \quad \square$$

On the chosen contours the integrand of Lemma 1 is absolutely integrable: on vertical lines with non-integral real part each factor $\pi/\sin$ decays like $e^{-\pi|\Im|}$ in its argument, so the triple decays like $e^{-\pi(|y_1|+|y_2|+|y_1+y_2|)} \leq e^{-2\pi \max(|y_1|, |y_2|)}$, while R_n has polynomial decay $O(|s|^{-2}|t|^{-2})$ in each variable. All the rearrangements below happen inside this absolutely convergent double integral.

3 Bivariate partial fractions of the kernel

Proposition 1. *The rational function R_n has poles exactly on the grid $s = s_k := -n-1-k$, $t = t_l := -n-1-l$, $0 \leq k, l \leq n$, of order at most two in each variable, and*

$$R_n(s, t) = \sum_{k, l=0}^n \left[\frac{C_{22}}{(s-s_k)^2(t-t_l)^2} + \frac{C_{12}}{(s-s_k)(t-t_l)^2} + \frac{C_{21}}{(s-s_k)^2(t-t_l)} + \frac{C_{11}}{(s-s_k)(t-t_l)} \right],$$

with local data (depending on n, k, l)

$$(-1)^n n! C_{22} = T(n, k, l), \quad \frac{C_{12}}{C_{22}} = L_k, \quad \frac{C_{21}}{C_{22}} = L_l, \quad \frac{C_{11}}{C_{22}} = L_k L_l - C_2.$$

Proof. For fixed generic t , $R_n(\cdot, t)$ is a proper rational function ($O(s^{-2})$ at infinity) with double poles at $s_0, \dots, s_n$ only, so it equals the sum of its principal parts; expanding each principal-part coefficient (a rational function of t of the same shape) in t gives the displayed grid decomposition with no polynomial remainder.

For the local data, write $s = s_k + x$, $t = t_l + y$ and factor the germ as $R_n = \Lambda F(x, y)/(x^2 y^2)$ with $F(0, 0) = 1$. Then $C_{22} = \Lambda$, $C_{12}/C_{22} = \partial_x F(0, 0)$, $C_{21}/C_{22} = \partial_y F(0, 0)$, $C_{11}/C_{22} = \partial_x \partial_y F(0, 0)$.

Leading coefficient. $\lim_{s \rightarrow s_k} (s - s_k)^2 \prod_{j=1}^n (s + j) / \prod_{j=n+1}^{2n+1} (s + j)^2$: the numerator at s_k is $\prod_{j=1}^n (j - n - 1 - k) = (-1)^n (n + k)! / k!$, while the surviving denominator factors are $\prod_{j \neq n+k+1} (j - n - 1 - k) = (-1)^k k! (n - k)!$, squared. Hence the s -factor contributes $(-1)^n (n + k)! / (k!^3 ((n - k)!)^2)$, and likewise in t . Finally $\prod_{j=n+2}^{2n+1} (s_k + t_l + j) = (-1)^n (n + k + l)! / (k + l)!$. Multiplying,

$$\Lambda = (-1)^n \frac{(n + k)! (n + l)! (n + k + l)!}{k!^3 l!^3 (k + l)! ((n - k)!)^2 ((n - l)!)^2} = \frac{(-1)^n}{n!} T(n, k, l).$$

First jets. $\partial_x \log F(0, 0)$ collects the logarithmic derivatives at $s = s_k$ of the three s -dependent products: $\sum_{j=1}^n \frac{1}{s_k + j} = -\sum_{i=k+1}^{k+n} \frac{1}{i} = -A_1(k)$; $-2 \sum_{j \neq n+k+1} \frac{1}{s_k + j} = -2(-H_k + H_{n-k}) = -2B_1(k)$; and $\sum_{j=n+2}^{2n+1} \frac{1}{s_k + t_l + j} = -\sum_{i=k+l+1}^{k+l+n} \frac{1}{i} = -C_1$. Hence $\partial_x F(0, 0) = -A_1(k) - 2B_1(k) - C_1 = L_k$, and symmetrically $\partial_y F(0, 0) = L_l$.

Mixed jet. Since only the coupled product depends on $s+t$, $\partial_x \partial_y \log F(0, 0) = -\sum_{j=n+2}^{2n+1} (s_k + t_l + j)^{-2} = -C_2$, so $\partial_x \partial_y F(0, 0) = L_k L_l - C_2$. $\square$

4 Recombination over the universal integrals

For integers $k_1, k_2 \geq 0$ and $s_1, s_2 \in \{1, 2\}$ define, following [1, Remark 1],

$$I_{k_1, k_2}^{(s_1, s_2)} = \frac{1}{(2\pi i)^2} \int_{\frac{1}{3}-i\infty}^{\frac{1}{3}+i\infty} \int_{\frac{1}{3}-i\infty}^{\frac{1}{3}+i\infty} \frac{1}{(u+k_1)^{s_1} (v+k_2)^{s_2}} \frac{\pi}{\sin \pi u} \frac{\pi}{\sin \pi v} \frac{\pi}{\sin \pi(u+v)} du dv. \quad (6)$$

Proposition 2 (recombination). *For every $n \geq 0$,*

$$I_n = \sum_{k, l=0}^n T(n, k, l) W_B(n, k, l), \quad W_B := I_{k, l}^{(2, 2)} + L_k I_{k, l}^{(1, 2)} + L_l I_{k, l}^{(2, 1)} + (L_k L_l - C_2) I_{k, l}^{(1, 1)}.$$

Proof. Insert the partial fractions of Proposition 1 into Lemma 1; absolute convergence permits integrating term by term. In the (k, l) term substitute $u = s + n + 1$, $v = t + n + 1$. The contours $\Re s = -n - \frac{2}{3}$, $\Re t = -n - \frac{2}{3}$ become $\Re u = \Re v = \frac{1}{3}$, no pole is crossed, and

$$\sin \pi s = (-1)^{n+1} \sin \pi u, \quad \sin \pi t = (-1)^{n+1} \sin \pi v, \quad \sin \pi(s + t) = \sin \pi(u + v),$$

so the sine triple picks up the factor $(-1)^{2(n+1)} = 1$, while $(s-s_k, t-t_l) = (u+k, v+l)$. Thus the (k, l) term contributes $C_{22}I_{k,l}^{(2,2)} + C_{12}I_{k,l}^{(1,2)} + C_{21}I_{k,l}^{(2,1)} + C_{11}I_{k,l}^{(1,1)}$, and by Proposition 1 the prefactor $(-1)^n n! C_{22} = T$ absorbs the $(-1)^n n!$ of Lemma 1. $\square$

5 Exact evaluation of the universal integrals

Brown–Zudilin [1, Remark 1] compute the Fourier–Mellin kernel

$$f(u, v) = \frac{1}{(2\pi i)^2} \iint u^{t_1} v^{t_2} \frac{\pi}{\sin \pi t_1} \frac{\pi}{\sin \pi t_2} \frac{\pi}{\sin \pi(t_1 + t_2)} dt_1 dt_2 = \begin{cases} \frac{uv \log u}{(1-u)(1-v)} - \frac{u \log(u/v)}{(1-u/v)(1-v)}, & 0 < u \\ \text{(same with } u \leftrightarrow v), & 0 < v \end{cases}$$

by residue analysis. Combining with the Laplace representation $\frac{1}{(t+a)^s} = \frac{1}{\Gamma(s)} \int_0^1 u^{t+a-1} (-\log u)^{s-1} du$ (valid on $\Re t = \frac{1}{3}$, $a \geq 0$) gives

$$I_{k_1, k_2}^{(s_1, s_2)} = \frac{(-1)^{s_1+s_2}}{\Gamma(s_1)\Gamma(s_2)} \iint_{[0,1]^2} u^{k_1} v^{k_2} f(u, v) (\log u)^{s_1-1} (\log v)^{s_2-1} \frac{du}{u} \frac{dv}{v}. \quad (7)$$

(The display in [1] omits the sign $(-1)^{s_1+s_2}$, i.e. writes $\log$ for $-\log$; the sign is invisible for $(s_1, s_2) = (2, 2)$ and $(1, 1)$ but reverses $(1, 2)$ and $(2, 1)$. It is fixed by the defining contour integral (6), and independently by direct numerical contour quadrature and by the calibration $I_{0,0}^{(2,2)} = I_0 = 2\zeta(5) + 4\zeta(2)\zeta(3)$.)

Expanding (7) on the two triangles by geometric series and integrating termwise produces an explicit evaluation. We record it in the form needed later. Define, for $x \geq 0$ and $r, m \geq 1$, the finite Euler sums

$$S_{r,m}(x) = \sum_{t=1}^x \frac{H_t^{(m)}}{t^r}, \quad U_{r,m}(a, b) = \sum_{t=1}^a \frac{H_{t+b}^{(m)}}{t^r}.$$

Proposition 3. *Each $I_{k,l}^{(s_1, s_2)}$ is a $\mathbb{Q}$ -linear combination of $1, \zeta(2), \zeta(3), \zeta(4), \zeta(5), \zeta(2)\zeta(3)$ whose coefficients are explicit polynomials in harmonic numbers at $k, l, k+l$ and the finite Euler sums above. The coefficients of weight $> s_1 + s_2 + 1$ vanish; in particular $\zeta(5)$ and $\zeta(2)\zeta(3)$ occur only in $I^{(2,2)}$, with the constant coefficients*

$$[\zeta(5)] I^{(2,2)} = 2, \quad [\zeta(2)\zeta(3)] I^{(2,2)} = 4.$$

The remaining coefficients are:

$$[\zeta(4)] : \quad I^{(1,2)} = I^{(2,1)} = \frac{17}{4}, \quad I^{(1,1)} = I^{(2,2)} = 0;$$

$$[\zeta(3)] : \quad I^{(1,1)} = 2, \quad I^{(1,2)} = 2(H_k - H_{k+l}), \quad I^{(2,1)} = 2(H_l - H_{k+l}), \\ I^{(2,2)} = 2(H_k^{(2)} + H_l^{(2)} - 2H_{k+l}^{(2)});$$

$$[\zeta(2)] : \quad I^{(1,1)} = H_k + H_l - 2H_{k+l}, \quad I^{(1,2)} = H_l^{(2)} - 2H_{k+l}^{(2)}, \quad I^{(2,1)} = H_k^{(2)} - 2H_{k+l}^{(2)}, \\ I^{(2,2)} = -4H_{k+l}^{(3)};$$

and the rational parts $r_{s_1 s_2} := [1] I_{k,l}^{(s_1, s_2)}$ are

$$r_{11} = (H_{k+l} - H_k - H_l)(H_k^{(2)} + H_l^{(2)}) - H_k^{(3)} - H_l^{(3)} + U_{1,2}(k, l) + U_{1,2}(l, k), \\ r_{12} = -2(H_k + H_l - H_{k+l})H_l^{(3)} + H_k^{(2)}H_{k+l}^{(2)} - \frac{1}{2}(H_l^{(2)})^2 + H_{k+l}^{(2)}H_l^{(2)} - \frac{5}{2}H_l^{(4)} + 2S_{1,3}(l) - U_{2,2}(k, l), \\ r_{21} = r_{12} \text{ with } k \leftrightarrow l, \\ r_{22} = -2(H_k^{(2)} + H_l^{(2)})(H_k^{(3)} + H_l^{(3)}) + 2H_{k+l}^{(3)}(H_k^{(2)} + H_l^{(2)}) + 2H_{k+l}^{(2)}(H_k^{(3)} + H_l^{(3)}) - 2H_k^{(5)} - 2H_l^{(5)} \\ - 6S_{1,4}(k) - 6S_{1,4}(l) - 2S_{2,3}(k) - 2S_{2,3}(l) + 6U_{1,4}(k, l) + 6U_{1,4}(l, k) + 2U_{2,3}(k, l) + 2U_{2,3}(l, k),$$

with $U_{r,m}(a,b)$ abbreviated $U_{r,m}$ evaluated at (a,b) .

Proof sketch, with the full reduction recorded. On the triangle $0 < u < v < 1$ substitute $u = xv$; both denominators expand geometrically, and every term is a product of two beta-type integrals in x and v carrying powers of $\log$. The termwise integration is justified by monotone convergence after splitting the logarithms into signs, or by integrating over $[\delta, 1 - \delta]^2$ and letting $\delta \rightarrow 0$; both triangle contributions have integrable boundary limits. The result is a bilinear combination of the tails

$$Z_m(N) = \sum_{t \geq N} t^{-m} \quad (m \geq 2), \quad Z_1 \text{ regularised}, \quad \sum_{t \geq 1} \frac{1}{t^r(t+h)^m}, \quad \sum_{t \geq 1} \frac{H_{t+d-1}^{(m)}}{t^r},$$

with $h \leq k_2$, $d = k_2 + 1$ and offsets governed by k_1 (and the mirror triangle swaps $(k_1, s_1) \leftrightarrow (k_2, s_2)$). The two divergent Z_1 -constants cancel pairwise ($a_1 + b_1 = 0$ in each partial-fraction splitting), so no regularisation ambiguity remains: every step is a limit of finite rational identities. Shifted products $\sum_t t^{-r}(t+h)^{-m}$ reduce by partial fractions to zeta values and harmonic numbers at h ; the coupled tails reduce to the double zeta values $\zeta(m, r) = \sum_{t > u \geq 1} t^{-m} u^{-r}$ plus the finite sums S, U ; and the weight-at-most-five double zetas reduce by

$$\begin{aligned} \zeta(2, 1) &= \zeta(3), \quad \zeta(3, 1) = \frac{\zeta(4)}{4}, \quad \zeta(2, 2) = \frac{3\zeta(4)}{4}, \quad \zeta(4, 1) = 2\zeta(5) - \zeta(2)\zeta(3), \\ \zeta(3, 2) &= 3\zeta(2)\zeta(3) - \frac{11}{2}\zeta(5), \quad \zeta(2, 3) = \frac{9}{2}\zeta(5) - 2\zeta(2)\zeta(3), \end{aligned}$$

together with $\sum_t H_t/t^2 = 2\zeta(3)$, $\sum_t H_t/t^3 = \frac{5}{4}\zeta(4)$ and $\zeta(2)^2 = \frac{5}{2}\zeta(4)$. Since the $\zeta(5)$ - and $\zeta(2)\zeta(3)$ -carrying heads $\zeta(4, 1), \zeta(3, 2), \zeta(2, 3)$ enter with (k_1, k_2) -independent coefficients, the top-weight coefficients are constants, fixed by the calibration $I_{0,0}^{(2,2)} = I_0$. Carrying out the finite bookkeeping for each (s_1, s_2) yields the displayed tables; the tables were additionally re-derived by an independent implementation of the same reduction and cross-checked in exact arithmetic on $0 \leq k, l < 14$ (both the derived and the displayed compact forms), so each entry is an identity of finite expressions whose verification is mechanical.

Two representative entries, worked in full, exhibit the bookkeeping. Write $a = k_1$, $b = k_2$, $A = a + 1$, $d = b + 1$; set $Z_m(N) = \sum_{t \geq N} t^{-m}$ for $m \geq 2$ and $Z_1(N) = -H_{N-1}$ (the regularised value; the divergent constants cancel pairwise), $U_{r,m}(h) = \sum_{t \geq 1} t^{-r}(t+h)^{-m}$, $S_{r,m}(A, d) = \sum_{t \geq A} t^{-r} Z_m(t+d)$, and $E_2 = \sum_t H_t/t^2 = 2\zeta(3)$, $E_3 = \sum_t H_t/t^3 = \frac{5}{4}\zeta(4)$ (these enter through $S_{r,1}$). Let $F_{a,b}^{s_1, s_2}$ denote the contribution of the triangle $0 < u < v < 1$, so that $I_{k_1, k_2}^{(s_1, s_2)} = (-1)^{s_1+s_2} [F_{a,b}^{s_1, s_2} + F_{b,a}^{s_2, s_1}] / (\Gamma(s_1)\Gamma(s_2))$ and

$$F_{a,b}^{s_1, s_2} = (-1)^{s_1+s_2-1} \sum_{i=0}^{s_1} \binom{s_1}{i} i! (s_1-i+s_2-1)! S_{i+1, s_1-i+s_2}(A, d) + (-1)^{s_1+s_2} \sum_{i=0}^{s_1-1} \binom{s_1-1}{i} (i+1)! (s_1+s_2-2-i)!$$

$[\zeta(4)]$ of $I^{(1,2)}$. For $(s_1, s_2) = (1, 2)$: $F^{1,2} = 2S_{1,3} + S_{2,2} - Z_2(A)Z_2(A+d-1)$, whose $\zeta(4)$ content is $2 \cdot \frac{1}{4} + \frac{3}{4} - \frac{5}{2} = -\frac{5}{4}$ (from $\zeta(3, 1)$, $\zeta(2, 2)$ and $\zeta(2)^2$; the finite sums $U_{r,m}(h)$ carry no $\zeta(4)$ since their zeta coefficients all come with positive powers of h^{-1}). For the mirror $(s_1, s_2) = (2, 1)$: $F^{2,1} = 2S_{1,3} + 2S_{2,2} + 2S_{3,1} - Z_2Z_2 - 2Z_3Z_1$, and $S_{3,1}$ contributes $-E_3 = -\frac{5}{4}\zeta(4)$; the $\zeta(4)$ content is $\frac{1}{2} + \frac{3}{2} - \frac{5}{2} - \frac{5}{2} = -3$. Hence $[\zeta(4)]I^{(1,2)} = (-1)^{1+2-2}(-\frac{5}{4} - 3)/(\Gamma(1)\Gamma(2)) = \frac{17}{4}$, as tabulated — note that without the Laplace sign the value would come out as $-\frac{17}{4}$.

$[\zeta(2)]$ of $I^{(2,2)}$. For $(s_1, s_2) = (2, 2)$: $F^{2,2} = -[6S_{1,4} + 4S_{2,3} + 2S_{3,2}] + 2Z_2(A)Z_3(A+d-1) + 2Z_3(A)Z_2(A+d-1)$. The isolated- $\zeta(2)$ content of the U -sums is $[\zeta(2)]U_{1,4}(h) = -h^{-3}$, $[\zeta(2)]U_{2,3}(h) = 3h^{-3}$, $[\zeta(2)]U_{3,2}(h) = -3h^{-3}$, so summing over $h \leq b$, $[\zeta(2)]S_{1,4} = H_b^{(3)}$, $[\zeta(2)]S_{2,3} = -3H_b^{(3)}$, $[\zeta(2)]S_{3,2} = 3H_b^{(3)} - H_a^{(3)}$ (the last term from $-\sum_{t \leq a} t^{-3}\zeta(2)$ inside the

coupled tail), while the Z -products contribute $-2H_{a+b}^{(3)} - 2H_a^{(3)}$. Altogether $[\zeta(2)]F_{a,b}^{2,2} = -[6 - 12 + 6]H_b^{(3)} + 2H_a^{(3)} - 2H_{a+b}^{(3)} - 2H_a^{(3)} = -2H_{a+b}^{(3)}$, and adding the mirror gives $[\zeta(2)]I^{(2,2)} = -4H_{k+l}^{(3)}$. $\square$

Remark 2. The finite Euler sums are not an artefact: the pair $(S_{r,m}, U_{r,m})$ is exactly the local trace the sine kernel leaves behind after the two-triangle integration, and the *basis* in which one writes them matters. The expressions above use the fixed basis {products of $H^{(\cdot)}$ at $k, l, k+l$; $S_{r,m}$ at k, l ; $U_{r,m}$ at $(k, l), (l, k)$ }; different-looking outputs of the same reduction are equal modulo the finite quasi-shuffle relations, e.g. $S_{r,m}(x) + S_{m,r}(x) = H_x^{(r)} H_x^{(m)} + H_x^{(r+m)}$ and $U_{r,m}(a, b) = H_a^{(r)} H_{a+b}^{(m)} - \sum_{j=1}^a H_{j-1}^{(r)}(b+j)^{-m}$.

6 The four kernel identities

Throughout this section we use the abbreviations $A_{kl} = C_{22}$, $B_{kl} = C_{12}$, $C_{kl} = C_{21}$, $D_{kl} = C_{11}$ of Proposition 1 (so $B_{kl} = A_{kl}L_k$ etc.), and the one-variable sections of $R := R_n(x - n - 1, y - n - 1)$, i.e. of

$$R(x, y) = \frac{P(x)P(y)P(x+y)}{Q(x)^2Q(y)^2}, \quad P(z) = \prod_{r=1}^n (z-r), \quad Q(z) = \prod_{r=0}^n (z+r),$$

whose partial fractions on the grid $x = -k$, $y = -l$ carry the same data A, B, C, D . For fixed l let

$$g_l(x) = \lim_{y \rightarrow -l} (y+l)^2 R(x, y) = \sum_{k=0}^n \left[\frac{A_{kl}}{(x+k)^2} + \frac{B_{kl}}{x+k} \right], \quad q_l(x) = \frac{\partial}{\partial y} \left[(y+l)^2 R(x, y) \right]_{y=-l} = \sum_{k=0}^n \left[\frac{C_{kl}}{(x+k)^2} + \frac{D_{kl}}{x+k} \right]$$

From the product form,

$$g_l(x) = c_l \frac{P(x)P(x-l)}{Q(x)^2}, \quad q_l(x) = \frac{P(x)}{Q(x)^2} [c'_l P(x-l) + c_l \lambda_l P(x-l)]', \quad (8)$$

more precisely $q_l(x) = P(x)Q(x)^{-2}(\mu_l P'(x-l) + \nu_l P(x-l))$ for finite constants μ_l, ν_l (the y -derivative never touches the factor $P(x)$). Consequently:

- (Z1) $g_l(j) = 0$ for $1 \leq j \leq n+l$ (both numerator products vanish on $1 \leq j \leq n$ resp. $l+1 \leq j \leq n+l$; the range is sharp);
- (Z2) g_l has a *double* zero on $l < j \leq n$, so also $g'_l(j) = 0$ there (sharp);
- (Z3) $q_l(j) = 0$ for $1 \leq j \leq n$ (the surviving factor $P(j) = 0$; sharp, and the zero has order exactly one).

Moreover $g_l, q_l = O(x^{-2})$ as $x \rightarrow \infty$, so the sums of their residues vanish:

$$\sum_{k=0}^n B_{kl} = 0, \quad \sum_{k=0}^n D_{kl} = 0 \quad \text{for each fixed } l, \quad (9)$$

and symmetrically with the roles of k and l exchanged ($A_{kl} = A_{lk}$, $C_{kl} = B_{lk}$, $D_{kl} = D_{lk}$).

Proposition 4 ($\zeta(5)$, $\zeta(2)\zeta(3)$, $\zeta(4)$). *For every n :*

$$[\zeta(5)] \sum_{k,l} T W_B = 2Q_n, \quad [\zeta(2)\zeta(3)] \sum_{k,l} T W_B = 4Q_n, \quad [\zeta(4)] \sum_{k,l} T W_B = 0.$$

Proof. The first two are immediate from Proposition 3 and (2): only $I^{(2,2)}$ carries these values, with constant coefficients 2 and 4.

For $\zeta(4)$: by Proposition 3 the coefficient is $\frac{17}{4} \sum_{k,l} T(L_k + L_l)$. For fixed l , $\sum_k T L_k$ is proportional to $\sum_k B_{kl} = 0$ by (9); summing over l , and treating L_l symmetrically, gives zero. $\square$

Proposition 5 ($\zeta(3)$). *For every n , $[\zeta(3)] \sum_{k,l} T W_B = 0$.*

Proof. By Proposition 3, up to the common factor $(-1)^n n!^{-1}$,

$$[\zeta(3)] \sum T W_B \propto 2 \sum_{k,l} \left[A_{kl} (H_k^{(2)} + H_l^{(2)} - 2H_{k+l}^{(2)}) + B_{kl} (H_k - H_{k+l}) + C_{kl} (H_l - H_{k+l}) + D_{kl} \right].$$

The D -term vanishes by (9). Using $H_k - H_{k+l} = -\sum_{j=1}^l \frac{1}{k+j}$, $H_k^{(2)} - H_{k+l}^{(2)} = -\sum_{j=1}^l \frac{1}{(k+j)^2}$ and the $k \leftrightarrow l$ symmetry to fold the $H_l^{(2)}$ - and C -halves onto the k -side, the bracket becomes

$$-2 \sum_{l=0}^n \sum_{j=1}^l \sum_{k=0}^n \left[\frac{A_{kl}}{(k+j)^2} + \frac{B_{kl}}{k+j} \right] = -2 \sum_{l=0}^n \sum_{j=1}^l g_l(j) = 0$$

by (Z1), since $1 \leq j \leq l \leq n+l$. $\square$

Proposition 6 ($\zeta(2)$). *For every n ,*

$$[\zeta(2)] \sum_{k,l} T W_B = -4 \sum_{k,l} T w_3^{\text{sym}}, \quad w_3^{\text{sym}} = \frac{1}{2} (H_{n+k}^{(3)} + H_{n+l}^{(3)}) - \frac{\Psi}{2} (H_{n+k}^{(2)} - H_{n+l}^{(2)}).$$

Proof. Step 1. By Proposition 3 (note $[\zeta(2)] I^{(2,2)} = -4H_{k+l}^{(3)}$), and again up to the common nonzero factor,

$$[\zeta(2)] \sum T W_B \propto \sum_{k,l} \left[-4A_{kl} H_{k+l}^{(3)} + B_{kl} (H_l^{(2)} - 2H_{k+l}^{(2)}) + C_{kl} (H_k^{(2)} - 2H_{k+l}^{(2)}) + D_{kl} (H_k + H_l - 2H_{k+l}) \right].$$

We first claim this equals $\sum_{k,l} [-4A_{kl} H_{k+l}^{(3)} - (B_{kl} + C_{kl}) H_{k+l}^{(2)}]$. The difference is

$$\sum_{k,l} \left[B_{kl} (H_l^{(2)} - H_{k+l}^{(2)}) + C_{kl} (H_k^{(2)} - H_{k+l}^{(2)}) + D_{kl} (H_k - H_{k+l}) + D_{kl} (H_l - H_{k+l}) \right],$$

and unfolding the harmonic differences into ranges as in Proposition 5, then folding the $(C, D_{(l\text{-side})})$ -half onto the k -side by symmetry, the difference equals

$$-2 \sum_{k=0}^n \sum_{j=1}^k \sum_{l=0}^n \left[\frac{B_{kl}}{(j+l)^2} + \frac{D_{kl}}{j+l} \right] = -2 \sum_k \sum_{j=1}^k q_k^{\text{mirror}}(j) = 0$$

by the mirror of (Z3) (the simple-pole-in- x section, as a function of y , vanishes at $y = j$ for $1 \leq j \leq n$, and $j \leq k \leq n$).

Step 2. It remains to identify $\sum T [H_{k+l}^{(3)} + \frac{1}{4} (L_k + L_l) H_{k+l}^{(2)}]$ with $\sum T w_3^{\text{sym}}$; multiplying by -4 then gives the claim. Multiply w_3^{sym} by $4A_{kl}$ and use $L_k - L_l = -2\Psi$ and symmetry: the difference of the two summands reduces to

$$\sum_{k,l} \left[4A_{kl} (H_{n+k}^{(3)} - H_{k+l}^{(3)}) + 2B_{kl} (H_{n+k}^{(2)} - H_{n+l}^{(2)} - H_{k+l}^{(2)}) \right]$$

after folding by symmetry. For fixed l the term $-2B_{kl}H_{n+l}^{(2)}$ sums to zero by (9). Expanding the two remaining harmonic differences over the range $k+l < i \leq n+k$, i.e. $i = k+j$ with $l < j \leq n$, leaves

$$\sum_{l=0}^n \sum_{j=l+1}^n \sum_{k=0}^n \left[\frac{4A_{kl}}{(k+j)^3} + \frac{2B_{kl}}{(k+j)^2} \right] = -2 \sum_{l=0}^n \sum_{j=l+1}^n g'_l(j) = 0$$

by (Z2). □

7 The middle row

This section proves Theorem 1. It is logically independent of Sections 4–6 except for reusing (Z1)–(Z2) and (9); that independence is what powers the subtraction argument of Section 8.

Brown–Zudilin’s descent [1, eq. (I3)], specialised to the totally symmetric parameters (write $k = n+l$ in their summation index), reads

$$2I''_n = \sum_{l=0}^n (-1)^l \binom{n+l}{n} \binom{n}{l}^2 J_3(n, n, n, n-l; n, n, n+l), \quad (10)$$

where J_3 is the generalised Beukers integral and $I''_n = Q_n \zeta(3) - \hat{P}_n$. (The factor 2 is discussed below; it is forced uniformly in n , not fitted.) Each term satisfies the Rhin–Viola condition $(n-l) + (n+l) = n+n$, so by [2, Sec. 3] it is a linear form $2A\zeta(3) - B$ whose coefficients come from the partial fractions of an explicit rational function: using Zudilin’s parameter correspondence ([2, eqs. (3.8), (3.10)]), the relevant function for the term l is

$$R_{n,l}(t) = \frac{\prod_{i=1}^n (t+i) \prod_{i=1}^n (t+i-l)}{\prod_{i=n+1}^{2n+1} (t+i)^2},$$

with double poles at $t = -n-1-k$, $0 \leq k \leq n$. Writing

$$R_{n,l}(t) = \sum_{k=0}^n \left[\frac{A_{kl}}{(t+n+1+k)^2} + \frac{B_{kl}}{t+n+1+k} \right],$$

the same product evaluations as in Proposition 1 give

$$A_{kl} = \binom{n+k}{n} \binom{n+k+l}{n} \binom{n}{k}^2, \quad B_{kl} = A_{kl} L_k. \quad (11)$$

Zudilin’s Lemma 4 [2, eq. (2.13) and its proof] evaluates the corresponding series as

$$J_3(n, n, n, n-l; n, n, n+l) = (-1)^l \left\{ 2 \sum_k A_{kl} \zeta(3) - \sum_k A_{kl} [2H_{k+l}^{(3)} + L_k H_{k+l}^{(2)}] \right\}.$$

Substituting into (10) and using that the outer coefficient times A_{kl} is exactly $T(n, k, l)$ (the six binomials merely reorder), we obtain the real-number identity

$$2Q_n \zeta(3) - 2\hat{P}_n = 2 \sum_{k,l} T \zeta(3) - \sum_{k,l} T [2H_{k+l}^{(3)} + L_k H_{k+l}^{(2)}]. \quad (12)$$

The $\zeta(3)$ terms on the two sides are *identically equal rationals* multiplied by the same real number $\zeta(3)$, by (2); subtracting them (no independence needed) leaves

$$\hat{P}_n = \sum_{k,l} T \left[H_{k+l}^{(3)} + \frac{1}{2} L_k H_{k+l}^{(2)} \right] = \sum_{k,l} T \left[H_{k+l}^{(3)} + \frac{1}{4} (L_k + L_l) H_{k+l}^{(2)} \right], \quad (13)$$

the second equality by $k \leftrightarrow l$ symmetry of T . Step 2 of Proposition 6 identifies the right side with $\sum T w_3^{\text{sym}}$. Finally w_3^{sym} is the symmetrisation of $\hat{w}_3$: their difference is $k \leftrightarrow l$ antisymmetric, and T is symmetric, so $\sum T w_3^{\text{sym}} = \sum T \hat{w}_3$. This proves Theorem 1, up to two normalisation points:

The factor 2 in (10). The iterated-residue derivation of (I3) in [1] produces the left-hand side as the residue integral displayed there; comparing its $\zeta(3)$ coefficient, computed by the right-hand side of (12) to be $2 \sum T = 2Q_n$, with the $\zeta(3)$ coefficient Q_n of I_n'' shows the displayed residue object equals $2I_n''$ for every n . Both coefficients are explicit rationals inside one derivation, so this comparison involves no independence statement. (At $n = 0$ it is the classical calibration: the residue integral is Beukers' $2\zeta(3)$ while $I_0'' = \zeta(3)$.)

I_n'' versus (1). The identity $I_n'' = Q_n \zeta(3) - \hat{P}_n$ is [1]'s definition of the middle-row pair via the same recursion-plus-initial-values argument as (1); its role here is only to name the rational number $\hat{P}_n$. $\square$

8 The subtraction anchor: proof of Theorem 2

By Propositions 2 and 3,

$$I_n = \sum_{k,l} T W_B = 2Q_n \zeta(5) + 4Q_n \zeta(2) \zeta(3) + 0 \cdot \zeta(4) + 0 \cdot \zeta(3) - 4 \left[\sum_{k,l} T w_3^{\text{sym}} \right] \zeta(2) + \sum_{k,l} T W^{(1)}, \quad (14)$$

using Propositions 4–6 for the five upper coefficients; here $W^{(1)} = [1]W_B$ is (3). On the other hand (1) gives

$$I_n = 2Q_n \zeta(5) + 4Q_n \zeta(2) \zeta(3) - 4\hat{P}_n \zeta(2) - 2P_n.$$

Subtract: by Theorem 1, $\sum T w_3^{\text{sym}} = \hat{P}_n$, so every transcendental term cancels *exactly*, leaving the equality of real numbers

$$\sum_{k,l} T W^{(1)} = -2P_n.$$

This is Theorem 2. $\square$

Remark 3. Theorem 2 is an unconditional closed form for the top row: P_n equals an explicit double sum of products of binomials, harmonic numbers, and the finite Euler sums $S_{r,m}, U_{r,m}$. It is the exact weight-five analogue of the middle row's (13), and it is the first representation of P_n obtained without any coefficient comparison.

Remark 4. The chain uses (1) and (2) from [1], whose proofs rest on a certified creative-telescoping recursion for I_n plus exact evaluation of I_0, I_1, I_2 ; an independent certificate for the same recursion, in normalised form, is part of the present programme's repository. No other external input enters.

9 The remaining bridge to the compact weight

Theorems 2 and 3 differ by the single identity (4), equivalently

$$\Delta_n := \sum_{k,l=0}^n T(n, k, l) [W^{(1)}(k, l) + 2w_5^{\text{sym}}(n, k, l)] = 0 \quad (n \geq 0).$$

This is a statement about finite sums only; no integrals or zeta values remain. We record its status precisely.

9.1 What is verified

$\Delta_n = 0$ has been checked in exact rational arithmetic for $n \leq 12$ (all cells, all six basis coefficients of the underlying decomposition compared independently), and modulo two 31-bit primes for $n \leq 80$. Moreover $P_n = \sum T w_5$ itself has been verified against the exact recursion ladder for $n \leq 360$ at two primes.

9.2 The proof strategy, and its measured state

All identities of Section 6 were proved by exhibiting the relevant weighted sum as a combination of *null functionals*: sums that vanish for every n because of (Z1)–(Z3), (9), the two-variable vanishing $R(x, y) = 0$ at lattice points and along the moving anti-diagonals $x + y = m$ ($1 \leq m \leq n$, all Laurent coefficients), and the one-variable residue lemma applied to $R_k(z)\rho(z)$ for admissible rational ρ (“pole-raising jets”, which carry the higher logarithmic jets $e_2, \dots, e_5$ of the local germ that fixed-pole evaluations cannot see). The natural certificate for (4) would be the same shape, in its *folded* form: since $B_{kl} = A_{kl}L_k$ and $D_{kl} = A_{kl}(L_kL_l - C_2)$ with L ’s polynomial in harmonic numbers, any cellwise identity

$$[W^{(1)} + 2w_5^{\text{sym}}](n, k, l) = \sum_i x_i u_i(n, k, l), \quad u_i \text{ null},$$

implies (4).

We record the measured state of this reduction (all statements modulo a 31-bit prime, with the protocol: solve on $n \leq N$, test consistency on the dependent rows, hold out $n = N+1$). With the generator inventory consisting of: weighted fixed-pole evaluations (including inverse-power and n -coupled weights), derivative evaluations on the double-zero range, anti-diagonal and vertical Laurent band families, pole-raising jets to order five, weighted value towers $\sum_{j \in \text{range}} \varphi(j) (R_k \rho)(-j) = 0$ with k -side letter multipliers (these produce the coupled sums $\sum_j \varphi(j)/(j+l)^s$ up to $s = 5$ that the U -letters of $W^{(1)}$ require), and two-variable iterated-residue jets $\sum_{k,l} \text{Res}_y \text{Res}_x [R \rho(x) \sigma(y)] = 0$ with lattice pole-raising in both variables (these turn out, measurably, to lie in the per-fibre span, so they enlarge nothing) — about 5,600 null functionals in all, each individually proved and machine-calibrated — the three-component system is consistent through $n \leq 14$ (with over a thousand dependent rows) but *inconsistent* at $n \leq 17$, and the folded system is inconsistent at $n \leq 20$ (514 of 3310 rows). Thus (4) is *not* a fixed-coefficient combination of these families, a bounded negative that sharpens the earlier codimension-one measurements. Two structural upgrades are identified and left open: coefficients rational in n (the certificate shape of every creative-telescoping identity in this area; the first augmentation test was row-limited and inconclusive), and *marginal* null weights $v(n, k) + v(n, l)$ with $\sum_k v(n, k) S_k(n) = 0$ for the fibre marginal $S_k(n) = \sum_l T(n, k, l)$, whose own residue calculus comes from the moment identities $\sum_l \text{Res}_{y=-l} [y^m R(x, y)] = -[\text{explicit rational}]$ — these are the only mechanisms that can carry the l -free nested content ($S_{1,4}(k)$, $S_{2,3}(k)$, ...) that the target provably contains and that no per-fibre or iterated-residue functional can produce monomially.

9.3 Why the compact weight is expected

The companion note *The Laurent-jet origin of the compact Brown–Zudilin weights* shows that $\hat{w}_3$ and w_5 are forced as Bell-polynomial coefficients of the anti-diagonal logarithmic jets of the local germ of R_n at its finite double poles — the direction that annihilates the coupled factor $P(x+y)$. Identity (4) is the statement that the same jets exhaust the rational content of the four universal integrals after T -weighted summation; every piece of $W^{(1)}$ ’s nested content lies in the image of the weighted-tower functionals, which is the structural reason the bridge is expected to close over the proved generator space.

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